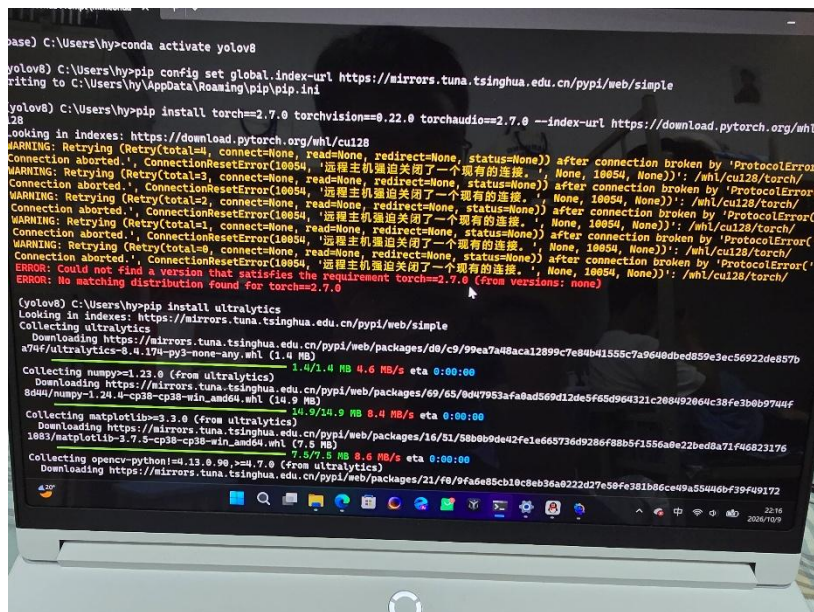
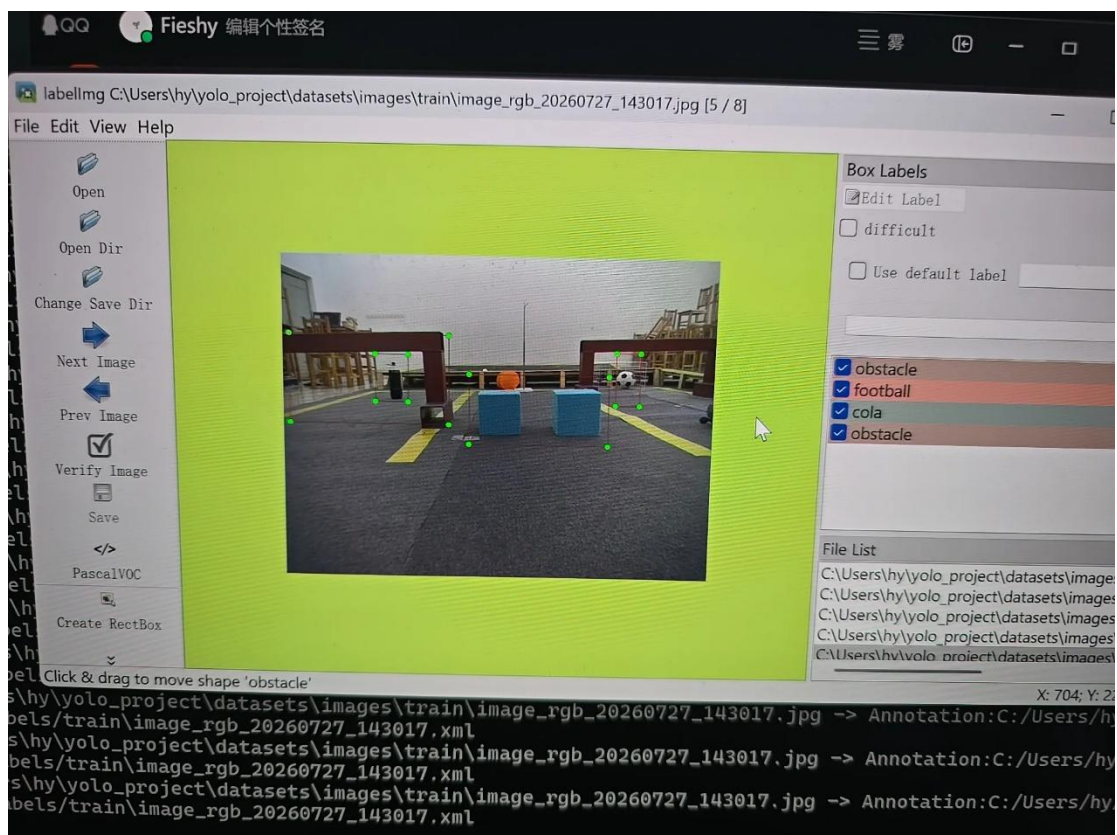


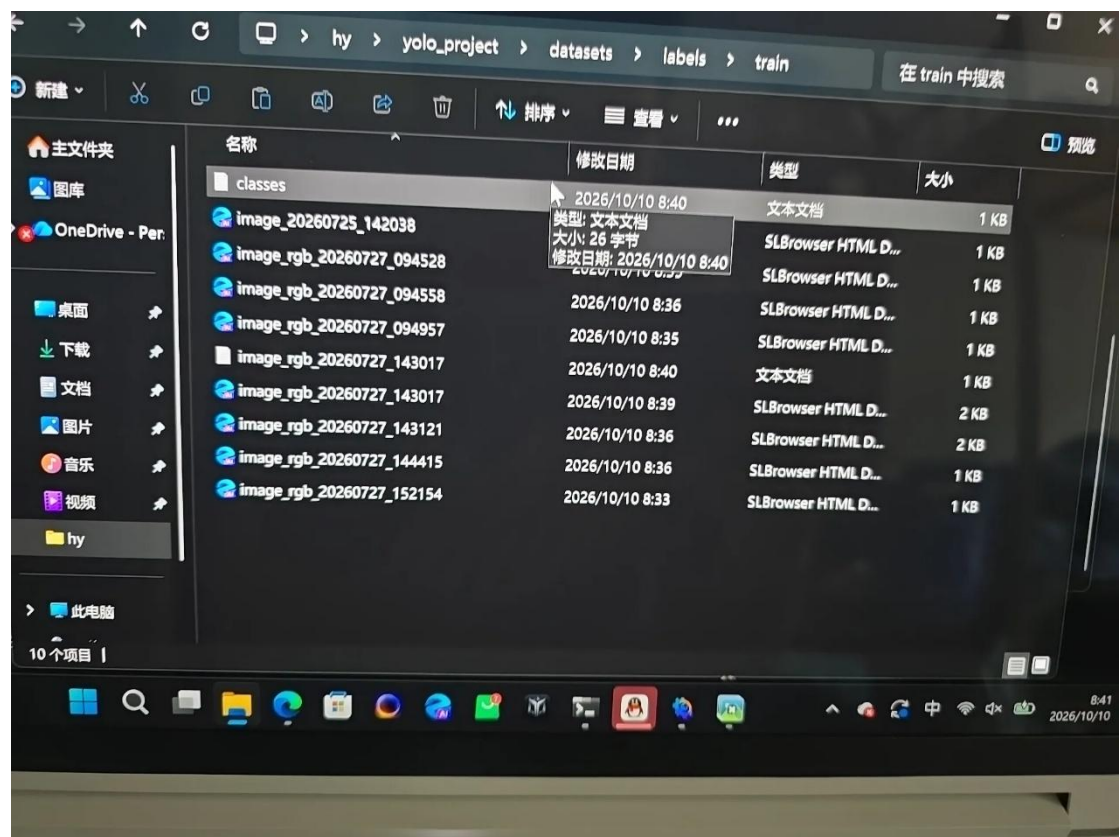
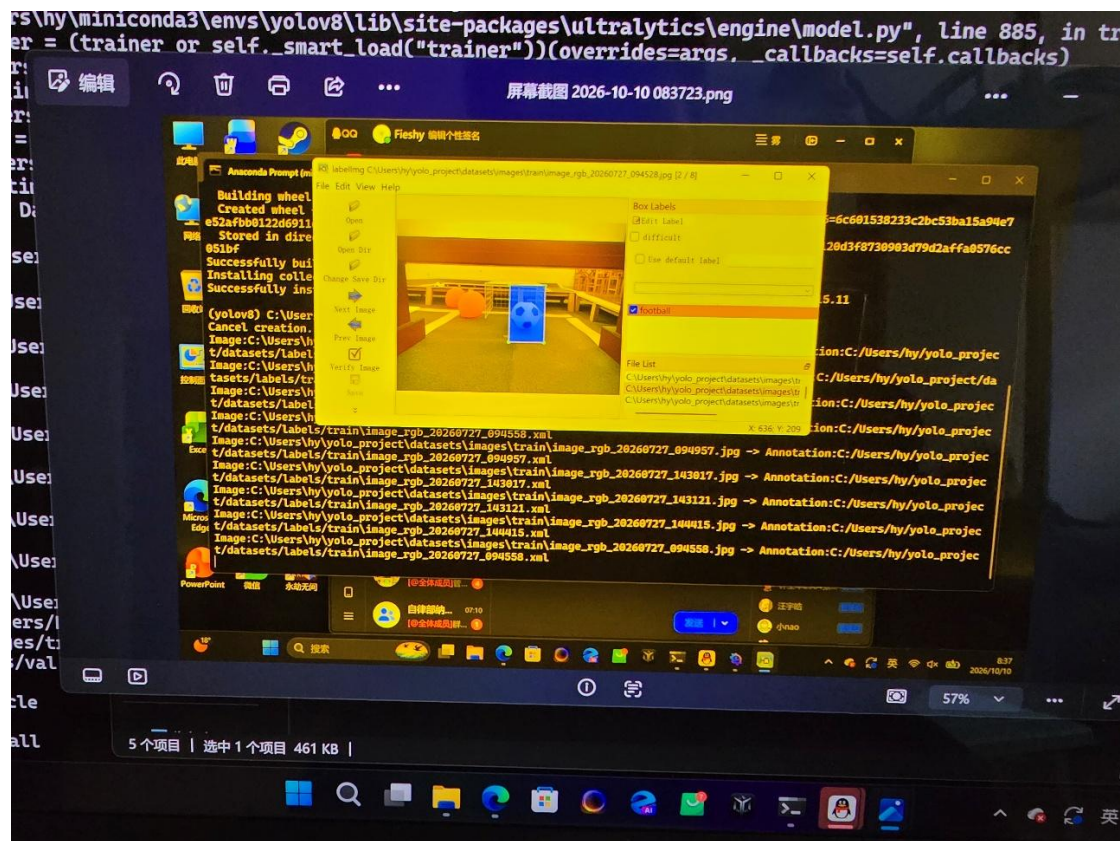
一、环境配置与网络排错 (Level 1)



刚开始查阅资料，发现出现 10060 和 10054 的报错，通过上网查阅发现，发现是国内网络受限，上面也显示了多次连接超时。我通过配置清华镜像源及挂加速器解决，成功完成环境搭建。也成功安装了 ultralytics。

二、数据整理与目标标注 (Level 2)

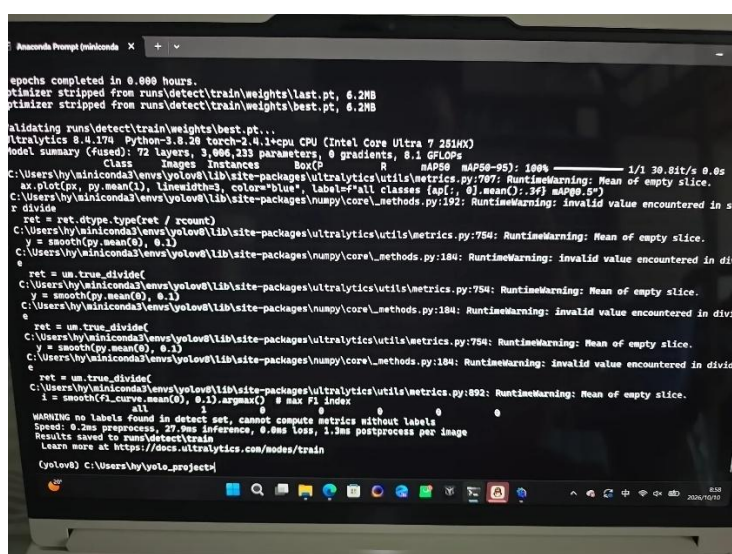




```
(yolov8) C:\Users\hy>cd yolo_project
(yolov8) C:\Users\hy\yolo_project>echo path: C:/Users/hy/yolo_project/datasets > data
(yolov8) C:\Users\hy\yolo_project>echo train: images/train >> data.yaml
(yolov8) C:\Users\hy\yolo_project>echo val: images/val >> data.yaml
(yolov8) C:\Users\hy\yolo_project>echo names: >> data.yaml
(yolov8) C:\Users\hy\yolo_project>echo 0: obstacle >> data.yaml
(yolov8) C:\Users\hy\yolo_project>echo 1: cola >> data.yaml
(yolov8) C:\Users\hy\yolo_project>echo 2: football >> data.yaml
(yolov8) C:\Users\hy\yolo_project>type data.yaml
path: C:/Users/hy/yolo_project/datasets
train: images/train
val: images/val
names:
  0: obstacle
  1: cola
  2: football
(yolov8) C:\Users\hy\yolo_project>yolo train data=C:/Users/hy/yolo_project/data.yaml mod
```

“下载实验室数据集，学习使用 Labellmg，刚开始是 xml 文件，然后用 ai 询问为什么这样，然后了解需要进行 YOLO 格式标注，然后成功创建 txt 的文件，然后按照要求构建了 images/train、val 与 labels/train、val 目录结构，并配置了 data.yaml 文件。

三、模型训练 (Level 3)



```
epochs completed in 0.000 hours.
optimizer stripped from runs\detect\train\weights\last.pt, 6.2MB
optimizer stripped from runs\detect\train\weights\best.pt, 6.2MB
validating runs\detect\train\weights\best.pt...
Ultralytics 8.4.174 Python-3.8.20 torch-2.4.1+cpu (Intel Core Ultra 7 251HX)
model summary (fused): 72 layers, 3,906,223 parameters, 0 gradients, 8.1 GFLOPs
Class      Images Instances BoxCp      A      mAP50  mAP50-95: 100% _____ 1/1 39.8it/s 0.0s
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\ultralytics\utils\metrics.py:707: RuntimeWarning: Mean of empty slice.
  ax.plot(px, py.mean(1), linewidth=3, color='blue', label='all classes [apf:, 0].mean():.3f mAP50.5")
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\numpy\core\_methods.py:192: RuntimeWarning: invalid value encountered in sc
r divide
  ret = ret.dtype.type(ret / rcount)
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\ultralytics\utils\metrics.py:754: RuntimeWarning: Mean of empty slice.
  y = smooth(py.mean(0), 0.1)
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\numpy\core\_methods.py:184: RuntimeWarning: invalid value encountered in divid
e
  ret = us.true_divide(
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\ultralytics\utils\metrics.py:754: RuntimeWarning: Mean of empty slice.
  y = smooth(py.mean(0), 0.1)
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\numpy\core\_methods.py:184: RuntimeWarning: invalid value encountered in divid
e
  ret = us.true_divide(
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\ultralytics\utils\metrics.py:754: RuntimeWarning: Mean of empty slice.
  i = smooth(F1_curve.mean(0), 0.1).argmax() # max F1 index
WARNING: no labels found in detect set, cannot compute metrics without labels
Speed: 0.2ms preprocess, 27.9ms inference, 0.0ms loss, 1.3ms postprocess per image
Results saved to runs\detect\train
Learn more at https://docs.ultralytics.com/modes/train
(yolov8) C:\Users\hy\yolo_project\
```

在 CPU 环境下完成 1 轮训练，成功生成 [best.pt](#) 权重文件。由于样本数量比较少且

算力有限，模型还没有完全熟练。

四、推理测试与对比反思（Level 4，我认为最重要的一环）

使用自训练模型推理时我未检测出目标。我分析了原因：可能是本次仅有 8 张训练图、CPU 训练 1 轮，数据量太少导致模型未熟练。为了验证推理环境，我使用官方预训练模型推理，成功检测出足球（置信度 0.94）。这让我深刻认识到数据量、算力对深度学习的重要性。虽然自训练模型未检出，但已完整跑通了从数据标注到推理的全流程。



```
epochs completed in 0.000 hours.
optimizer stripped from runs\detect\train\weights\last.pt, 6.2MB
optimizer stripped from runs\detect\train\weights\best.pt, 6.2MB

validating runs\detect\train\weights\best.pt...
ultralytics 8.4.170 Python-3.8.20 torch-2.4.1+cpu CPU (Intel Core Ultra 7 251HX)
Model summary (fused): 72 layers, 3,006,233 parameters, 0 gradients, 8.1 GFLOPs
Class Images Instances Box(P R MAP50 MAP50-95): 100% 1/1 30.8it/s 0.0s
ax.plot(px, py.mean(1), linewidth=3, color='blue', label=f'all classes (ap: {0} mean(): 36f mAP@0.5s)
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\numpy\core\_methods.py:192: RuntimeWarning: invalid value encountered in scalar divide
r divide
ret = ret.dtype.type(ret / rcount)
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\ultralytics\utils\metrics.py:754: RuntimeWarning: Mean of empty slice.
y = smooth(py.mean(0), 0.1)
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\numpy\core\_methods.py:184: RuntimeWarning: invalid value encountered in divide
ret = un.true_divide(
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\ultralytics\utils\metrics.py:754: RuntimeWarning: Mean of empty slice.
y = smooth(py.mean(0), 0.1)
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\numpy\core\_methods.py:184: RuntimeWarning: invalid value encountered in divide
ret = un.true_divide(
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\ultralytics\utils\metrics.py:754: RuntimeWarning: Mean of empty slice.
y = smooth(py.mean(0), 0.1)
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\numpy\core\_methods.py:184: RuntimeWarning: invalid value encountered in divide
ret = un.true_divide(
C:\Users\hy\miniconda3\envs\yolov8\lib\site-packages\ultralytics\utils\metrics.py:892: RuntimeWarning: Mean of empty slice.
i = smooth(f1_curve.mean(0), 0.1).argmax() # max F1 index
WARNING no labels found in detect set, cannot compute metrics without labels
Speed: 0.2ms preprocess, 27.0ms inference, 0.0ms loss, 1.3ms postprocess per image
Results saved to runs\detect\train
Learn more at https://docs.ultralytics.com/modes/train
(yolov8) C:\Users\hy\yolo_project\
```